

DEMONSTRATION OF PROPOSED FEATURES

OptiCrop: Smart Agricultural Production Optimization Engine

Date	03 July 2026
Team ID	SWTID-2026-7253
Team Size	5
Team Leader	Bommidi Deepika
Team Members	G Uday Kumar, Kavali Anu, Pavan Kumar Sowdepalli, Dasarayyagari Sravan Kumar
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Marks

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated[cite: 42]. It serves as evidence of the team's ability to deliver on the proposed solution[cite: 42].

Feature Implementation and Demonstration Matrix

S.No	Feature Name	Description	Status	Demo'd (Yes / No)	Remarks
1	User Management	Secure user registration and login with authentication[cite: 42].	Implemented[cite: 42]	Yes[cite: 42]	Functional as per PEP guidelines[cite: 42].
2	Applicant Form Workflow	Interactive submission form collecting regional soil and environmental parameters[cite: 42].	Implemented[cite: 42]	Yes[cite: 42]	Includes real-time input validation[cite: 42].
3	Data Preprocessing Module	Handles data cleaning, normalization, label encoding, and feature scaling[cite: 42].	Implemented[cite: 42]	Yes[cite: 42]	Using Pandas and Scikit-learn[cite: 42].
4	ML Prediction Engine	Processes applicant data through trained ML models for agricultural crop optimization prediction[cite: 42].	Implemented[cite: 42]	Yes[cite: 42]	Generates status and confidence metrics[cite: 42].
5	Database Integration	Secure storage and retrieval of application records and prediction history[cite: 42].	Implemented[cite: 42]	Yes[cite: 42]	Using MySQL database[cite: 42].
6	Admin Dashboard	Administrator monitoring of system performance, predictions, and statistics[cite: 42].	Implemented[cite: 42]	Yes[cite: 42]	Analytics and report generation[cite: 42].

Feature Implementation Summary

Metric Category	Value
Total Features Proposed	6[cite: 42]
Total Features Implemented	6[cite: 42]
Total Features Demonstrated	6[cite: 42]
Overall Implementation Rate	100%[cite: 42]